Karl Friston & Alexander Ororbia: “First Day ~ Symposium Roundtable Discussion“

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is Carl hello welcome back everyone I'm here with Alex aoria we may have Carl join we may have someone else join if you want to join maybe write it in the YouTube live chat and maybe I'll email you a link otherwise write some questions however to kick things off Alex feel free to say hello and start in and I know that there's things that we can explore together sure you just picked it back to me and I was saying to you all right well I I'll just introduce myself because I'm not entirely sure how most of the audience if they know of who I am so I'm Alex Ria I am a computational neuroscientist and cognitive scientist who has worked for active inactive inference and specifically predictive coding versions of it for quite some time I'm Carl's friend and collaborator uh some of you might be aware of more of our recent arguments of mortal computation which is a Cory or a new coraly or another coraly of the free energy principle and uh yeah and you know I'm all about what's wrong with deep learning in machine intelligence and I'm usually the guy you go to if you want to attack back propagation of Errors large language models Transformers kind of relates to a lot of the discussion I was able to catch Snippets of earlier today so if you want me to help dismantle what's what's the problem with deep learning you know that that can be fun too so that at least if nothing else everyone now knows who I am Carl needs no introduction maybe start off with that while I try to figure out why he cannot join oh okay uh what part of that do you want me to I mean I can go into Manifesto but I'm not sure that's what this panel should be about out maybe beginning with a five minute Manifesto and then there will be some questions and uh we'll see where we get well maybe to try to make it useful to the audience uh for the moment although I wasn't intending the panel to be focused about that um maybe one thing that we can talk about is there was this discussion during the active inference Workshop about metabolism and sort of instill a homeostatic metabolic drive a survival almost orientation to machine intelligence and so uh Carl and I have now been arguing for probably now about more than a year since uh our paper came out like on yeah around November actually start of November talking about uh another way to redesign reformulate to use Ma's earlier uh uh wording uh intelligence itself uh particularly machine intelligence and so mortal computation without again if for those that might not be familiar actually my friend Jeff also like two years ago introduced this phrase the idea that computation or intelligent calculations and substrate uh in biological natural systems are entangled and it's very hard to divorce those two concepts uh whereas computer science and a lot of like what cat GPT is or large language models or your favorite GPT structure is Immortal which is the idea that we can copy that program including its weight tensors and it's encoded knowledge and place it on GPU server you want and it'll behave roughly the same there's nothing inherent about its physical instantiation uh in this universe uh that dictates or shapes or uh necessarily embodies it right and so there's this like move to uh embodiment by the way active in the free energy principle has all of these characteristics mortal computation which is again as the concept I explain earlier Carl and I sort of uh recast it we strongly redid it and said that um uh because while Jeff didn't want to necessarily directly explore it we put it as it's just basically thinking of the free energy principle and taking it to its natural extreme conclusion which is that machine intelligence systems uh because they are inherently bound to their substrate must focus on this survival objective right and the free energy principle and one of its ways to talk about it uh basically is saying you know we have a markup blanket or a nested system of marup blankets and our goal in this universe is to protect those internal States uh yet interact vicariously with our external States and so that eventually gives us our ultimate goal which is to prevent ourselves from disintegration into a heat bath it's a very physics uh form of thinking about identity and identity construction um but this is kind of what's centrally missing in basically all of machine intelligence um is this the completely right path to go there's plenty of criticisms and I'm not one to say that there's only one direction we need to go if we ever want something like artificial general intelligence um but this might be a path worth exploring it also addresses the Energy Efficiency arguments I was hearing earlier discussed uh Transformers particular l in large language models are uh grossly negligent of their energy costs they often times have the carbon footprint of a large city and that's actually now out of date as you know Carl and I wrote that about a year ago um there's even discussions I find kind of humorous by individuals I'm not going to particularly name saying we should be powering now uh these Transformers and large language models with nuclear power plants and so we're kind of going down the rabbit hole of let's just burn as much of our resources as we can to just get this particular type of intelligence I argue very strongly narrow AI uh and we're cheating ourselves of looking and embracing actually Energy Efficiency uh which would lead us towards the real grand goal of green AI uh because clal GL global climate change is clearly and empirically a thing and we should be a little worried about that so uh mortal computation sort of helps steer everything into sort of what I would argue hopefully maybe in multiple years time the emergence of a whole new field because I consider at least my personal take I'd love to have you know heard Carl wherever he might be um but I Envision actually mortal computation is not just like the framework and definitions that Carl and I talked about which kind of take free energy uh and active inference to the strongest conclusion to a naturalistic machine intelligence maybe perhaps a new domain of thought of biomedic intelligence and where everyone else can come together and play in this really interesting space and there's a lot to unpack there but I don't think we want to make this about that uh there's plenty of talks I've given including active inference Institute we've had a podcast actually I think it was right when the paper came out so for those that are interested I just recommend you to go to the YouTube channel and watch wonderful videos that Dan LED oh is there is there a new link okay sure was was I just blabbing there to no one no that was perfect we we we just hot swapped okay I was trying to buy Carl welcome Carl we talking about you okay thank you for the overview on Mortal computation and and and it was a highlight of the year um Carl with the evening Sunset now how um goes it wait I don't hear you oh oh continue sorry it's all good Carl continue can you hear me now yep excellent yeah I'm sorry about that I got locked out I was uh paying the price for enjoying myself for the first two hours right I'll wait just I really enjoyed the um the intervening sessions um and look forward to of comenting some the cross cutting scen yes okay um Alex mute it's a little noisier on jitsy and then uh corl um also a little quiet but we hear you so um what were the what were the themes that that uh connected the presentations that you were in or otherwise one thing that um struck me can you hear me all right now is this y excellent I mean there are lots of lots of little uh golden nuggets um throughout all the presentations um but one um one crosscutting theme was this notion of community that emerged in various guises such as the notion of Commons that was the subject of uh a lot of the exchanges in the first um in the first um uh presentation um and then the notion of community that was a Dumont of the the workshop um and then the not of coherence and synchronization and narrative social um science accounts of that and how that might be formalized in terms of um joint free energy minimization ensembles of Agents you know finding a coherent communal um and empowered free energy Minima so you know I thought I I thought that that was a common theme sort of focusing much less upon um individual agents but what happens when you have a an ecosystem of

Agents um and what principles um can be brought to the table to understand interactions or ensembles of Agents so I I have a number of you know lots of things came to mind I was trying as a physicist to sort of um rewrite all of these wonderful Concepts um in a mathematical way or or a physics way um so for examp examp I was think about the importance of Commons and it struck me that one way that we could motivate the choice of capital c for prior preferences uh would be that these are the common goals and the and the C characteristics States um that are shared between um multiple agents uh and of course um the last presentation took us through some of the mechanisms and the capital c circular capital c causality um that leads to this Capital C coherence so you know which originally was meant to be constraints in the sense of a constraint Maxim entry principle um or cost in the you in a reinforcement learning uh reinforcement learning setting but coming back of course all this started when I was hearing about Commons and how it underwrote communities it it is very interesting how development and the presentations have proceeded a lot more about the communication and somewhat of the multi-agent coordination as opposed to a longer memory train for individuals or deeper prospection for one individual um and in a way the renormalization method brought a a Clarity or a rigor to what had already been qualitatively explored from 2013 and Life as We Know It And all these other works that we're talking about multiscale systems and about blankets of blankets and within this year to make it timely we start to see some of the ways in which that compositionality can actually be brought into the generative models that we've seen I mean it looks like the textbook figure 4.3 and the nested models and now that's actually the motif that gives a way of looking at a multiplicity at the more inactive level as having a Unity at a higher level Alex in your designing or work how how does the single agent or multi-agent design come into play now yeah yeah can hear okay good yeah I was having mic troubles earlier um well so it's kind of interesting I think I've spent most of my career at the agent level in the sense that I want to build I don't think we have right the single agent and I and I think it's kind of important to sort of scope that out from an architectural learning perspective I'm very much a fan of like Carl's arguments and you know I Echo them myself a lot which is the idea of the inference learning and structural time scales uh I still think we have a lot to do but if I were to say how in my research the multi-agent approach really comes into play well I do think a lot in terms of assemblies or ensembles right putting together multiple circuits I'm a I'm a huge proponent of another type of this comes from cognitive science called the common model of cognition uh I'm not sure how many in the audience are familiar with it but it's sort of like a generic reprint that has emerged over the last several decades of cognitive science to say that we can think about and take our current Knowledge from neurop physiology and cognitive studies that brain has these rough levels of organization different types of memory we have the motor cortex we have phas of gangway we have all these very important structures and we can start appending to them some functionality and so in that sense that's where I think of multiple agents right because you can start breaking apart using again Carl and I've have argued this the mean field approximation uh there's natural mean field approximations that emerg through the result of ution where you have these little substructures that communicate with each other um obviously the sparse connectivity that Carl mentioned and his talk this morning is uh kind of indicative of how these little substructures emerge and then communicate and perform sort of like what uh Marvin Minsky would have said you know sort of a society of mind right uh so it's kind of a call back to that and I think in that regard I often uh one branch of my work often tries to highlight and emphasize bring forth that notion of cognitive Blueprinting or cognitive structure um because then we can start filling in the pieces with what we know about neurobiology like for example I use predictive coding quite a bit uh you know to fill in those gaps and so I think in that sense I think a lot of interacting subsystems you could think of those as agents and then my final comment uh without letting myself blab too much uh I have worked in neuroevolution so there is another scale Above This which is you know you do need to account for how multiple mortal computers or multiple agents interact with each other um and so I think in that regard I have done some work with very simplified agents where you can kind of abstract away the neuroscience and just say I want to look at these little tiny systems that interact and then we can you know modify them through genetic algorithms and neuroevolution so there is a little bit of that like higher level uh societal kind of learning I have looked at that and I think a big challenge just to bring it to the audience for the panel is uh merging those time scales in a in a coherent and efficient way I still think well there has been good work I know Daniel you work on and Colony optimization as well and other types of kind of aggregate algorithms uh I think bringing those time scales together and really interacting with what we currently know about single agents even like the common model of cognition and going back to the highest level which is this multi-agent evolutionary time scale is a Grand Challenge and how do we of course effectively simulate those in a way that anyone can play with this I think is a big Challenge and yeah and I think I'll kind of stop there before I go down more radicals neuroevolution and neural Darwinism something Carl and and others have also explored it's like they're all and then the multiple nested time scales multiple nested spatial scales it's like there's this attractor to want to come back and find the unifying patterns and and it's and how those can be applied is kind of an exciting question I'll read one question in the chat so feel free to give a thought or take it any direction David Highland wrote do you view mortal and IM do you view mortal and Immortal computation as binary categories or are they more like two points along a spectrum actually want Carl to answer that I mean I could give something but I really want to hear your thoughts on that it's your favorite subject though design I know but I'm actually really curious if you think of it as a spect because remember the conversation we've had with Mike Levan and Chris fields and we've been sort of debating you know how important it is to lean into mortality versus immortality do you think that there could possibly be anything in between because I honestly am you know I think of it kind of more rigidly I'll just give my aners that and I firmly think we need to like focus on Mortal computation but do you think that they're can be a hybrid in between so sorry to turn it on you but I'm very curious to know your thoughts um my my instinctual answer is no um having said that I am sure that you can present some um description of mortal computation in terms of immortal algorithms I mean I'm just thinking back to sort of um um the trilogy due to David Mah uh but does that mean that these kinds of descriptions are apt um for simulating reproducing understanding um or describing the actual process I I I suspect not they may at one level of description they may be but when it comes to um applying the principles usually in computer simulations I suspect that that just would not work I mean because Daniel and I were wrestling with zoom and other um um communication things I I I may we may have missed whether you've taken people through the the the key distinction between Mortal and Immortal uh computation so Al do you just want to briefly rehearse the sort of the two Cardinal views of the of the distinction um between Mortal and Immortal uh computation I did I just wasn't sure if anyone heard me so I wasn't sure what was going on Zoom yeah I mean certainly from the point of view the free energy principle the free energy principle is a description of a random dynamical system um and as such um it is um a system that constitutes the world the you know the lived World um perhaps not the lived world but certainly a mortal world um that has um characteristic time scales and therefore um the free energy principle and everything that ensues from that including active inference um rests upon that primary assumption that you are describing a process a mortal process a a a a process that could have um a physics um although um the actual free energy principle itself is almost pref physics so you have to derive basic mechanics and classical mechanics and quantum mechanics from um the existence the your the the the mortality of this process cast in terms of your time involving systems so in that sense um you know there is no such thing as Immortal computation but I also I don't know if you said it out loud but I also liked your um more playful take on Mortal versus Immortal that if so if software was Immortal uh and it had some autopoetic aspect to it then it wouldn't care about dying so there would be no wait there' be no way there' be no incentive to have characteristic States coming back to another big sea um uh so you know in that sense they cannot be immortal computation simply because that is removing the definitional uh stipulative definitional aspect of the characteristic states that we are trying to describe through the feny principle and active inference and specific um VAR variance of that such as predictive coding but I just pick up on this crosscutting CC theme uh you also um um celebrated that with common cognitive models so again with we've got this this s notion commonality and um and you asked the question you how How would how would one um apply these Notions and if you like sort of naturalize the common cognitive model under um a less anthropomorphic or functionalist active inference account and you gave the answer implicit in your question which how would you simulate these things um I think Daniel asked the same the same question indeed the the whole point of this Symposium is about applications so I I just wanted to put a question to the two of you I was um impressed with the very um clear distinction between agent-based modeling and the common cognitive equivalent the where the agents the members of The Ensemble were actually for example active inference agents that seems to me a very big move and a very difficult thing to actually do um most of the literature I know in this area um has at most taken baby steps into having ensembles of two agents with authentic agency in the sense that they can plan so I'm now limiting um the notion of an agent Beyond a thermostat or a virus to those kinds of systems that um have in mind or part of their Gena model the consequences of their Lear actions so they can be future pointing and they can plan and select from alternative policies uh and of course the numerics and the you know the the Practical difficulties of actually simulating doing agent-based modeling for example um with authentic agents in the mix in the ecosystem um I think is you know um is a really important Challenge and a really important distinction between classical agent-based models where you put lots of ther stats or rents tractors together and see what happens and of course they have a synchronization manifold and you can certainly wax lyrical about um generalized synchrony and and and Joint free energy minimization of a generic sort but I think that misses the point I think Ma was making um um that you know part of the the multiple L Loops of circular causality in sort of your know cultural Niche construction for example um you know really does rest upon um the capacity of any individual within an ensemble um to um act given they have selected a particular course of action particular path into the future which they have beliefs about the consequences of and you know to me that that that that is an authentic kind of agent so you know mean perhaps well both of you d mean your your your ethology background um are you aware of anybody's actually sort of Bitten the bullet that was introduced to US during the workshop and actually started creating active inference agents that actually plan and learn from each other and carve out their own Wallington Landscapes you know finding those sort of joint joint Minima I don't know specific uh citation but certainly you've given a research Direction and it reminded me of in the shared protentions work where thinking about what what does a shared Ensemble what would it be for a planning Ensemble or for a collective it could be a shared protention as in there's the same plan disseminated across the ant nestmates it could be that they have different maybe nonoverlapping but complimentary parts that would be shared it could be having a shared reference but they could have different plans or it could even be a shared consequence without even a shared reference so there's so many ways in which especially when we consider ensembles that that are heterogenous and they have different kinds of umelt different kinds of ways of engaging with the niche and and possibly some comment niche as well as some uniquely accessible components there's so many flavors of Ensemble that I I can can only hopelessly or helpfully try to think about what tools and infrastructure and coordination would even give us a grasp to be able to sweep or design or visualize across that otherwise we might be locked in way too early to what it means for The Ensemble to have a collective protention and then that's not going to apply to every ant species and if it can't even apply to all the ants out there then it's not going to work for aliens Alex can I can I just respond to that and then ask ask for your your view on that one so I think it's a really important point and um one of the big takeaways for me on our paper on the variational synthesis you you're applying the renormalization group to the the coupling of evolutionary to um sort of phenotypic uh time scales was that it was not uh feasible or appropriate to talk about consp specifics in isolation that it was you had to have the all the varieties of predators and praise and things that we um that not even animate not even agents in the mix to actually um derive the um your the non-equilibrium steady States at particular at particular scales which you know speaks directly it is not um it is not uh useful to um you know if you like take the idealized gas mentality of 20th century physics and now apply it to agent-based modeling to assume that all the agents are exchangeable and are identical in some s sense and then are converging on some common narrative um or some common um uh is not the right way to you know to simulate these things you can't do Predator prey relationships you know for example you can't have um even simple economic games uh but furthermore you you you when people talk about depletion of resources you actually have to model the resource you know whether it's oxygen food or water as things in in that ecosystem so I think that's a really important observation which just speaks to the challenge of using numerical studies to test out some of the hypotheses that uh that we' you know being being exposed to um you know you know if one just pursues agent-based modeling without introducing that heterogeneity I think one is making exactly the same mistake um that people make when they apply uh equilibrium physics to non-equilibrium systems um you know you know a true gas is not an ideal gas and it only has the kinds of properties and breaks detailed balance and expresses non-ra in virtue of the fact that there is no exchangeability you can't assume that every atom or every agent in our context um is exchangeable with with every other agent and to connect again to Commons and and that model of well within the firm we assume Co cooperativity and hierarchy in the market we have a flatness and a competitiveness and then there's this Commons that has sort of elements of each so then I think about doing ecosystem modeling it'd be like well we have within the ant colony we assume cooperativity possibly hierarchy among ant colonies you have red and tooth and Claw and competition and then it's like but can there be a unifying framework that would have the two genotypes and two phenotypes and two species and two adjacent niches and keep on adding in these levels of Distinction and seeing all those different variables in their joint Evolution across multiple time scales rather than the concatenation of a Cooperative hierarchical game NE Within in a lateral competitive game it's like capturing what those models get the the width and the height of but bringing it into a framework where it's being seen as just one joint evolving process so C is the letter of the day uh Alex can you speak to CES of cells what what that kind of heterogeneity and structured diversity would look like from a point of view of tissue for example or an organ okay wait what was the c word you said I just didn't hear it it's any any word starting with a letter C you can use but you have to oh I have to use okay this is the game um Community I'm gonna pick that one just because it starts with c um so okay Carl threw in organs and organel but I'm going to push that aside for just a minute to address I think one of the questions I heard emerge from yours and Carl's discussion which was uh is there anything like what you are arguing for which is avoiding the mistakes and by the way Predator prey model came immediately to your speaking Carl um about these classical models that assumed interchangeable agents and is there anything that looks at these heterogenous systems or heterogenous agent interactions and a couple of thoughts I don't think there's anything like directly what we would want and I think that ties or centers around our even the arguments you and I and many others in the community are making about uh why it's difficult to show case why active inference is so important even for places like machine intelligence um but I did write some thoughts quickly uh so I do think uh the first thing that came to mind weirdly enough was an application in the world of machine intelligence where they had some environment where they had a community of even though I say it with slight distaste large language models or Chad gpts or gpts that had some extra modality equipped to them and they had this like little Community now it was a virtual very simplified Niche if you will um but the idea is that they could observe this community um and so these different gpts would and I can try to dig up the paper and send it to Daniel later it's one of my students that actually pointed it out to me they would different roles they had different personalities if you will um and I know that they investigated that to some degree now of course that doesn't have any active inference in there and I'm not sure what degree of maybe perhaps light reinforcement learning being introduced but there is something like that and there is slight baby steps if you will uh in investigating some notion of embodi multi-mod and agent systems I think the AI Community is still sort of barely starting to embrace embodiment let alone enactivism and let alone embeddedness which is the idea of a community to go back to the seword of multiple agents interacting with each other um and investigating how do social norms and merge or cultural norms and cultural constraints and relationships at this like more Global level so I think there is a little bit of those Sparks of that thought and obviously I might not be aware of further follow-ups another place that uh I at least see the potential of no longer having these interchangeable particle like agents is in multi-agent robotics um I know that they have done things like soccer competitions and they have these little and again each of those robots they are the embodiment of what you and I want which is a version of a mortal computer so it's a little like assembly of mortal computers and there's a Cooperative nature so and then a competitive nature so you get a little bit of a very constrained real physical real world Niche um obviously it's in the constraints of it's a soccer game or some very simple task so I know in multi-agent robotics there is some development along this path and it'd be wonderful to have like a a a hardcore because I'm more of a simulation roboticist a hardcore roboticist to talk to us about the state of multi-agent Robotics I think that's the place where you could begin to test uh hypotheses and kind of or theories that are constructed even from active inference point of view uh into these real world robotic multi- assemblies you know they have like we could analyze the competition

Cooperative Dynamics I think that's our best bet in terms of what maybe exists at the closest to literal agents that aren't just very simplified even though they make simplifications in robotics too uh in order to have a little brain in there that you know does something meaningful usually learning is turned off which is my own criticism of Robotics um and I think the last comment I wanted to make and again getting back to the Notions of what we need so again you and I have a fondness I think for some of cybernetics and cybernetics has this notion of complex emergent systems and we you know you have these little uh building blocks at one level that follow their own physics or their own physical laws and then they maybe assemble or emerge into these higher level building blocks that are subject to their own laws then you have bottom up in top down causation which is a notion that you know comes from cybernetics and we always talk about that but then I think about like Santa you you might be familiar with the Santa Fe Institute and all their wonderful work like uh what was it not just Herbert Simon but uh uh John Holland and all the wonderful work on complex adaptive systems but they always did at the end of the day when they simulated it they did the particle thing this interchangeable particle simulation and then tried to analyze like those convergence properties or those emergent rhythms or patterns that kind of emerge in those systems with no to my knowledge and you know and I did study complex adaptive systems a while ago uh not with that heterogenity and let alone any of the complex agents that we could begin to design today even with deep learning regardless of its limitations and so I think that that maybe even turns into an answer to your question highlights an open opportunity Maybe revisiting and recasting Notions of cybernetics and complex adaptive systems in this idea of well can we actually build more complex agents uh for example that actually have neural substrate structures or something that we can simulate and then learn to acquire their own characteristics in their Niche I think the reason that maybe a lot of people avoid it or maybe why I don't perceive uh development in that direction is the problem also of Niche and you and I also had an and a not to try to highlight this paper but in the appendix we talked about the body Niche problem uh and that is this issue that uh you need a niche if you don't have a niche and you're designing these agents independently like deep learning typically does to apply to any Niche uh you're missing really half the problem which is understanding the physical laws and how do we simulate them such that people can then play with those models maybe you don't have access to a xenobot like Michael Levan does or I don't have access to organizes though I wish I did um but you know maybe we can simulate these and so I think that we need the community may this is a call to the community to develop uh viable niches virtual morphologies virtual uh areas that we can then connect these agents and then simulate many of them and try to analyze and understand do the things we predict even with these interchangeable particle models apply to these heterogenous communities that might emerge and again I guess that just to wrap up and I promise I'm done you brought up organel and organs and I think that highlights something that I wish I had more easy access to rather than uh being put into this position where I well may I need to build it which is you know these virtual organ systems or virtual organ organel and trying to understand how these dynamical systems interact and then I can kind of take that as that's my body or that's where my Mark and I can kind of start to put some instantiation of blankets and then say Okay I want to instill maybe a neural circuit maybe I put in a predictive coding model into this kind of virtual morphology so I can analyze and understand does do these interactions of the time scills actually come through when we actually build you know physical instantiations of these models and so I think that's going to be one of the biggest roadblocks to even my dreams of mortal computation which is this notion of myology something that we can actually work with that it gets as close as we can to reality without reality and in Niche um because once you have those problems solved then you can start building multi and if you have efficient way to simulate them and I'm sure lots of people that are watching might be worried uh because it's very hard to simulate you know real physical systems let alone allow machine intelligence researchers to play with them so I blabbed a little bit there but does that sort of give you some something to bite on based on your questions something to chew on something to consider um in the last little bit I'll I'll ask some questions so I'm going to read one from the live chat and then anyone else we'll just do some quick fun random questions sort of like sampling from the uncertainties expressed uncertainties of our live chat colleagues okay Pablo FM wrote is playing a simulation acquiring knowledge SL experiences through simulation of scenarios and how can we use active inference to do games how do you fellows make your active inference work playful and fun if or when let you go Carl you go I'm not sure if there are two there there are two questions there um a practical answer to um why you might want to instantiate um a simulation environment with active inference agents or indeed just a single agent in the in the spirit of computational phenotyping but I think probably more interesting in the um in the context of um communities of Agents um is in the spirit of this sort of cognitive um agent based modeling um is the ability to um parameterize the priors of the agents to best explain or replicate Legacy data from a particular ecosystem be it Financial epidemiological meteorological um purely ethological um and having optimized the parameters you then roll out into the future and in rolling out into the future you now have the opportunity to have a grounded in principle base optimal plus inactiv ISM and embedded um um estimate of what is likely to happen in the future again in the spirit of weather forecasting but of course this becomes much more um if like useful um when you have control of the system so Financial Market forecasting or epidemiological forecasting or uh climate change forecasting you assuming that there's a substantive um an um anthropomorphic um um anthropological um contribution to um climate and you all the other factors that uh you know um pertain to um resilience and uh resistance to things he you know health and the like um as soon as you can intervene then once you've got a digital twin of this ecosystem that has been optimized in relation to um the past data in the sense that this now has the highest model evidence or marginal likelihood in relation to that Legacy data you've then got an opportunity to do scenario modeling with interventions so you can now ask questions what would happen if we um cut this Supply CH train chain what would happen if we impose this lockdown in this viral pandemic what would happen and so on and so forth furthermore because you've got um um uncertainty in the mix because this is effectively going to be um you're using variation procedures that we use in active inference to actually simulate um the behavior that best matches the um the Legacy data you've also got for free uncertainty quantifications and not only can you say if I interv on this system and did this um then we would uh I can predict that in six months time or in six years time that and I can give you Bas in credibil intervals o over that moreover you can now apply the principles of active inference to roll out under different interventions which now become policies and evaluate the expected free energy of each policy and select the one under the constraints the Comm the characteristic outcomes U that you um commit to at this point in time and you can now use this scheme to make recommendations on the basis of the um the relative probability of this sequence of interventions versus that sequence of interventions so I think there are great practical importances of being able to put these principles in silico and basically do Evolution very very quickly so that tomorrow you can read the you can read the consequences probabilistically uh of evolution in this setting and that setting where the setting is actually something that we can intervene on or even if we can't just to know the consequences of our Collective Behavior Uh today so I think that would be um that's why you might want to make you know to invest in in these applications um with the fun and uh having uh and play I think that's a different issue I think that's basically um um one way of looking at the um the way that authentic agents behave under under the free energy principle which is basically to explore and gather the right kind of evidence that uh provides literally evidence for their generative models of of their world um which basically entails novelty seeking but now complement this with this um basing reduction we've been talking about also trying to find the minimal explanation for the data and the evidence that has been secured by exploratory playful Behavior arriving at particular insights which uh those are har moments that say oh I see it's just one of those or this one thing explains all of this playfully AC acquired data and perhaps that kind of mechanism you know may be a really important aspect of the communal um um you cognitive agent based or yeah cognitive agent based I can't quite remember the acronym it was very nicely laid out in one of the I have to go back and look at the live stream a follow-up question from the live chat Arno wrote I agree with the approach to calibrate to pass data and forecast but it's already quite challenging for current models that only have a few parameters to be calibrated how then does one calibrate prior beliefs of millions of Agents I'm talking about epidemiology and economic models sorry I'm going to give a technical answer before Alex gives a gives aedy answer uh you do it it's only difficult because 99% of agent-based modeling uh uses the sample distribution um you do not do that you use variational you actually um simulate and roll out the um the um probability distributions themselves using variational calculus and that means that what was yesterday um really difficult agent-based modeling and certainly epidemiological modeling now becomes really trivial uh orders of magnitude more uh more efficient what you have to take care of is exactly the thing that Daniel was was highlighting before which is a heterogeneity you need to put that into the mix and this manifests in a number of ways um so when you apply um variational procedures to effectively um roll out population density uh probability density distributions over various states of being am I infected am I staying at home um have I died um you have to make sure that the heterogeneity is part of the generative model so this is sometimes called stratification so for example in epidemiological models you'd have eight bands for different age groups and if you notice the way that the epidemiological data was generated was usually age segregated so that people could stratify their models um another um way in which this heterogeneity becomes absolutely crucial um when you come to um agent-based modeling in the context of epidemiological or infectious disease modeling is called over dispersion now over dispersion is the fact that uh you get heavy tailed distributions that inherit from the heterogeneity so you get this for free once you put heterogenity into the model but it does mean um for every if you like subgroup of um the population you do induce more parameters but remember the number of parameters it doesn't really matter it's the amount of time that you take to effectively invert your model and if you try and do that using sampling based approaches which is currently the the norm you could take literally weeks to uh do something um whereas if you use variational procedures then you could do that within within minutes so it's a really important question a very practical one yeah to restate direct variational inference on beliefs about ensembles themselves being variational belief inference on agent level semantic beliefs rather than sample from the sample what the agents are sampling yeah you're you're you're invoking the The Twist yes how on Earth can you model a variational density over things that have variational densities you can do it under the f p room because the only observable things about an agent are on its Markoff blanket and of course they are not belief structures so there is a lawful relationship between you can um if you like simulate cognitive agents using traditional agent-based uh approaches but what you have to do is to amortize or to or to um um basically work out the um the mapping between the history of what something sees um and what it actually does but in principle you can do it by marginalizing out the um the beliefs of the agent in principle I'm not and that's that's why I asked that's why I'm so intrigued by this distinction because that will be a challenge Alex and then Carl just in our last minutes here what would you say was a favorite memory or moment in theory and practice for active inference this year and then what are you excited about for next year um I'll go I'll go first so a favorite memory of active inference let me think quickly well I I mean I'm going to be a little by just to give a highlight and a shout out and everyone will get to share in that memory if you stick around after the round table is actually some of the results from one of my PhD students um via and he's going to give a talk soon and uh the idea is that uh one of the things I was proud of is he's been working in the area of Robotics which I think is a really good playground uh is difficult and hard is it to democratize for everyone to work with robotics I think it really shows you how far we need to go and I think that's a good place to make strong advances an active inference um and one of my memories is that I remember him kind of showing me uh because he's working with partially observable Markoff decision processes and how difficult and unaddressed it is in the field of working with raw sensory data to try to construct these continuous models and I remember him showing me some of his models rolled out trajectories in the future and I I'm always kind of in a fondest in my heart for generative models and I remember him playing out sort of these robotic memories of this robot trying to you know fantasize where it would want to go to kind of like move this block or to knock a ball towards a go so that was one of those moments where we were really proud because the idea is we actually we actually made a breakthrough uh in actually showing that we can construct a very robust framework that applied across all these robotic control problem and so I was immensely proud of that and so I think my students drive a lot of some of my fond memories um and uh and then we got uh Chris Buckley shout out to him uh to you know uh kind of give us his insights into our work too and it was a really really nice work and so and I think the best way to say it is stick around if you want to know how that all works I was really proud of that so one of my favorite memories of the Year awesome Carl with the last words of a fun memory and uh a direction or an excitement for next year right I'm I'm not going to tell you about my most fun moments that's that's very personal but my my my my excitement for next year is to see whether any of your hairs go gray after you do the master ceremony of these of not only these uh Symposium all the meetings because for those people who don't know Daniel has to run this almost singlehandedly nonstop for a marathon sometimes 10 hours at a time I don't know how he does it at Le unless you're using hair dye you're you're you're standing up physically remarkably well sir so I just see how long that lasts thank you Carl I look forward to the marathon changing continuing and flourishing in the coming year we appreciate you Daniel thank you guys it's been awesome okay thanks a lot see you soon nice talking to you guys bye everyone e e